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The Holonomy Theorem: Admissibility and the Exact Corrected Family $\tau^*(\kappa) = 27/(2\kappa)$

Diagonal or dead · holonomy-invariant station ladder · Berry closure +0.062%

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Abstract

This monograph records the holonomy theorem of the corrected system: the C1-ring residues vanish if and only if the two zero-channel constants coincide ($\kappa_{C1} = \kappa_{C2}$), the branch residues vanish only at $\kappa_{C2} = \kappa_{TH}$, and the κ -free doors audit shows no leaks — hence a holonomy correction is admissible only as a uniform renormalization $\kappa \rightarrow \kappa_{hol}$. The corrected clock family is exact, $\tau^*(\kappa) = 27/(2\kappa)$, with the holonomy-invariant station ladder $R3h = 3/2$, $W2h = 9$, $R5h = 183/40$, $M5h = -105/4$, $P4h = 153\sqrt{3}/10$, $P2h = \sqrt{3}/2$, $T0h = 3\sqrt{3}/2$ and the ring normalization $W2h/T0h^2 = 2\kappa/3$.

The book of the family is $\kappa_{book} = \ln(27/4) = 4\ln(3/2) + \ln(4/3)$ (deviation -2.574% against the observed anchor); the additive Berry screening $\kappa = 2 - \pi^2/98$ closes it to $+0.062\%$ without fitting, while the monograph-sign multiplicative book moves away (-7.43%). The spectrum at τ^* factors as $9\lambda^4 \cdot Q_{12}(\lambda)$ in $Q(\sqrt{3})$: the only genuine real root is $\lambda = 0$ (multiplicity four), all complex roots are phantoms, and the monodromy per echo is exactly 1. Tables, corollaries, honest caveats and a full reproduction protocol are included.

Keywords: holonomy, admissibility, corrected clock family, station ladder, Berry screening, monodromy, SymPy, machine theorem.

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1. Introduction

This monograph opens the English block of the Twelve Theorems cycle and records the holonomy theorem of the corrected system: the machine-proved admissibility statement for spinor corrections and the exact corrected clock family $\tau^*(\kappa) = 27/(2\kappa)$ with its holonomy-invariant station ladder. The result was delivered by campaign v11 (sympy_spectrum_tau_star.py together with sympy_spinor_corrections.py, LEDGER §22) and independently confirmed by the one-brick Berry machine s4_berry_one_brick.py.

The theorem answers three questions at once. First, how may a holonomy-type correction enter the tower without killing the CSS point: the ring consistency requires $\kappa_{C1} = \kappa_{C2}$ (the two zero channels must be corrected equally) and the branch consistency requires $\kappa_{C2} = \kappa_{TH}$, so the correction is admissible only as a uniform renormalization $\kappa \rightarrow \kappa_{hol}$. Second, what survives the correction: the exact family $\tau^*(\kappa) = 27/(2\kappa)$ and the station ladder $R3h = 3/2$, $W2h = 9$, $R5h = 183/40$, $M5h = -105/4$ are holonomy-invariant. Third, which book closes the observed clock: the additive Berry screening $\kappa = 2 - \pi^2/98$ moves $\ln(\tau^*)$ from -2.57% to $+0.062\%$ of the observed value — a measurement-grade closure without any fitting.

2. Notation and Setting

The corrected system carries a dimensionless constant κ entering the books through the ring normalization: the ring clock reads $W2h/T0h^2 = 2\kappa/3$. Three κ -channels are distinguished: κ_{C1} and κ_{C2} — the constants of the two C1-ring zero channels — and κ_{TH} — the constant of the TH branch. The static chain, the books UV_ξ3 and Mdef_ξ5, and the critical point x^* are as in the Russian block of the series (monographs No. 1–6); the reader is assumed familiar with the chain forms $R3h = 2R1h \cdot T0h^2/9$, $W2h = 4T0h^2/3$, $P2h = T0h/3$.

The setting of the theorem is an audit of correction schemes: the machine embeds a one-parameter family of holonomy corrections into the tower and asks when the CSS point survives. Twelve schemes were tested; ten keep the point alive, two kill it (the family scheme and the deliberate violation demo). The admissibility statements below are proved on the surviving sector.

Table 7.1. Notation used in this monograph

Symbol	Meaning
κ_{C1}, κ_{C2}	correction constants of the two C1-ring zero channels
κ_{TH}	correction constant of the TH branch
κ_{hol}	the uniform (diagonal) holonomy constant — the only admissible form
$\tau^*(\kappa) = 27/(2\kappa)$	exact corrected clock family
$b_C = \pi^2/98$	the Berry spinor quantum of the surviving brick $\delta_C = \pi/7$
κ_{obs}	the observed constant, $\kappa_{obs} = \gamma_{lit} \cdot \ln 10 / \Delta_{sp}$ (comparison anchor)

3. Statement of the Theorem

Theorem 7 (holonomy admissibility and the exact corrected family). Let the corrected system be embedded with channel constants $\kappa_{C1}, \kappa_{C2}, \kappa_{TH}$. Then:

(i) the C1-ring residues vanish if and only if $\kappa_{C1} = \kappa_{C2}$: the holonomy correction must be diagonal, one constant for both zero channels; any asymmetric embedding kills the CSS point (machine fact of the scheme audit);

(ii) the branch consistency $\kappa_{C2} = \kappa_{TH}$ collapses the three channels to one: the correction is admissible only as a uniform renormalization $\kappa \rightarrow \kappa_{hol}$, and no κ -free “leak” exists in the SC/UV/Mdef geometry (the doors audit: no leaks — there is no error in the equations to be corrected);

(iii) the corrected clock family is exact:

$$\dot{\iota}^{\dot{\iota}}(\dot{\iota})\dot{\iota}\{27\frac{\dot{\iota}}{2\dot{\iota}}, \dot{\iota}\dot{\iota}\dot{\iota}\dot{\iota}\dot{\iota}\kappaappa_{\dot{\iota}\{hol\}}, \quad (7.1)$$

with the holonomy-invariant station ladder $R3h = 3/2$ (exactly), $W2h = 9$, $R5h = 183/40$, $M5h = -105/4$, $P4h = 153\sqrt{3}/10$, $P2h = \sqrt{3}/2 = \sin(\pi/3)$, $T0h = 3\sqrt{3}/2$, and the ring normalization $W2h/T0h^2 = 2\kappa/3$;

(iv) the book of the family is $\kappa_{book} = \ln(27/4) = 3\ln 3 - 2\ln 2 = 4\ln(3/2) + \ln(4/3)$ (five books plus one zero pivot over six stations); its deviation from the observed anchor is -2.574% ; the additive Berry screening $\kappa = 2 - \pi^2/98$ closes it to $+0.062\%$, while the monograph-sign multiplicative book moves away (-7.43%).

4. Sketch of the Machine Proof

4.1. Ring consistency (i)

The machine embeds the corrections into the C1-ring residues and simplifies. The residues factorize, and every surviving factor carries the difference $\kappa_{C1} - \kappa_{C2}$: the residues vanish identically in κ exactly when the two channel constants coincide. This is proved as a polynomial identity in $Q(\kappa_{C1}, \kappa_{C2}, \kappa_{TH})$, so the conclusion “diagonal or dead” is exact, not a numerical observation. The scheme audit demonstrates the contrapositive: asymmetric embeddings move the CSS point off the constraint set, and the two dead schemes record the mechanism.

4.2. Branch consistency and the doors audit (ii)

The second layer embeds the TH branch and shows the branch residues vanish only at $\kappa_{C2} = \kappa_{TH}$; combined with (i), all three channels collapse to a single κ_{hol} . The doors audit checks the κ -free geometry: the SC/UV/Mdef doors close identically with and without κ , i.e., the tower does not “leak” — there is no pre-existing error that a correction would fix. Consequently the correction can only renormalize, never repair; this honesty point guides the whole embedding programme.

4.3. The family, the ladder, and the books (iii)–(iv)

With κ uniform, the corrected chain equations solve exactly, and the clock factor $2\kappa \cdot T0h^2 - 27$ has the unique positive root $T0h = \sqrt{27/(2\kappa)}$, whence (7.1). The station ladder is invariant: $R3h = 3/2$, $W2h = 9$, $R5h = 183/40$, $M5h = -105/4$ do not depend on κ — they are holonomy-blind; only the clock moves. The log-closed ladder has monodromy product exactly 1 around the six-station loop (log-sum 1.2×10^{-125}). The book decomposition and the Berry closure are quoted from s4_berry_one_brick and the v12 one-brick run: $\ln(\tau^*)$ closes from -2.574% (baseline) to $+0.062\%$ (brick).

Table 7.2. Layers of the verdict (spinor_corrections / spectrum_tau_star)

Layer	Statement	Status
S1 doors	SC/UV/Mdef geometry is κ -free; no leaks	exact

Layer	Statement	Status
S2 ring	C1-residues vanish iff $\kappa_{C1} = \kappa_{C2}$ (diagonal)	exact (polynomial)
S3 branch	$\kappa_{C2} = \kappa_{TH}$ required; uniform κ_{hol} only	exact (polynomial)
spectrum	at τ^* : char = $9\lambda^4 \cdot Q12(\lambda)$, deg 16; only genuine real root $\lambda = 0$	exact + $Q(\sqrt{3})$
books	$\kappa_{book} = \ln(27/4) = 4\ln(3/2) + \ln(4/3)$; Berry closes to +0.062%	exact form + measurement

5. Numerical Results

Table 7.3. The holonomy-invariant station ladder at x^*

Station	Exact value	Decimal
T0h	$3\sqrt{3}/2$	2.5980762114
R3h	$3/2$	1.5
W2h	9	9.0
P2h	$\sqrt{3}/2 = \sin(\pi/3)$	0.8660254038
R5h	$183/40$	4.575
M5h	$-105/4$	-26.25
P4h	$153\sqrt{3}/10$	26.4970402652

Table 7.4. The clock book and its closures (ln-measure against κ_{obs})

Book	Form	Deviation
Baseline κ_{book}	$\ln(27/4) = 3\ln 3 - 2\ln 2 = 4\ln(3/2) + \ln(4/3)$	-2.574%
Berry one-brick	$\kappa = 2 - \pi^2/98$ (additive screening, bC sector)	+0.062%
Monograph-sign	multiplicative book (opposite sign)	-7.43%

The spectrum at τ^* completes the picture: the characteristic polynomial of the corrected prolonged system has degree 16 and factors as $9\lambda^4 \cdot Q12(\lambda)$ in exact arithmetic over $Q(\sqrt{3})$; the only genuine real spectral point is $\lambda = 0$ (multiplicity four), and all twelve complex roots are phantoms (full pencil rank at each, $\text{rel } \sigma_{\min} \approx 10^{-4}$ – 10^{-6}). The point τ^* is an isolated

algebraic vertex of the constraint landscape, not an exponential repeller; the monodromy per echo is exactly 1 — the cycle physics lives in the finite-amplitude DSS limit, not in the point.

6. Corollaries

Corollary 7.1 (one brick moves the clock to the observed value). The Berry screening $\kappa = 2 - \pi^2/98$ is the additive book of the surviving spinor sector; combined with the exact family (7.1) it closes $\ln(\tau^*)$ to +0.062% of the observed anchor without fitting. The selection status of δ_C itself is governed by the structural barrier theorem (monograph No. 10) — the closure is an observation, not a selection principle (multiple-testing caveat of v18a).

Corollary 7.2 (invariance constrains embeddings). Any future correction scheme must preserve the ladder $R3h = 3/2$, $W2h = 9$, $R5h = 183/40$, $M5h = -105/4$ and move only the clock along (7.1). Schemes that move the ladder are outside the admissible class by (i)–(ii) — a falsifiable design rule for the embedding programme.

Corollary 7.3 (brick-ladder compatibility). Setting $\kappa(\delta) = 2 - \delta^2/2$ turns (7.1) into the brick-ladder law $\tau^*(\delta) = 27/(4 - \delta^2)$ of monograph No. 2: the two theorems are one statement in two parametrizations, which is why the two-point protocol of the repository is legitimate.

7. Honest Caveats

The admissibility proof is exact for the tested correction class (diagonal channel renormalizations embedded into the C1-ring and TH-branch residues); broader classes (non-diagonal, amplitude-dependent) were not exhausted — the scheme audit covers twelve specific embeddings. The Berry closure is a measurement against the anchor κ_{obs} built from $\gamma_{\text{lit}} = 0.374$ and $\Delta_{\text{sp}} = 7\pi/30$; anchors are labelled as such and nothing is fitted. The sign convention between additive and multiplicative books remains open (−7.43% versus +0.062%); the monograph-sign choice moves away from the observation and is recorded for the record. Finally, the spectrum verdict describes the truncated tower; the O7 programme (question No. 5 of OPEN9) may refine τ^* , though the exact identity of monograph No. 8 (two independent routes to one number) makes a substantial shift unlikely.

8. Reproducibility

Table 7.5. Reproduction protocol for Theorem 7

Step	Command	Output
1	<code>python3 sympy_spectrum_tau_star.py</code>	results/spectrum_tau_star.json (family + ladder + spectrum)
2	<code>python3 sympy_spinor_corrections.py</code>	results/spinor_corrections.json (ring/branch consistency)
3	<code>python3 s4_berry_one_brick.py</code>	one-brick Berry screening $\kappa = 2 - \pi^2/98$
4	<code>python3 -m pytest tests/ -q</code>	58 passed (holonomy tests included)

All machines are deterministic and single-threaded (BLAS = 1); runtimes are minutes on a laptop and comparable in Termux. The data correspond to commit 5b256cd; the self-contained machine copies live in `einstein_direct/open9/machines/` with a private results copy.

References

- [1] Repository wild8highlander/choptuik_ac_bc: `einstein_direct/sympy_spectrum_tau_star.py`; `einstein_direct/results/spectrum_tau_star.json` (campaign v11, LEDGER §22).
- [2] Ibid.: `einstein_direct/sympy_spinor_corrections.py`; `einstein_direct/results/spinor_corrections.json` (ring consistency, scheme audit).
- [3] Ibid.: `einstein_direct/s4_berry_one_brick.py` (Berry one-brick screening $\kappa = 2 - \pi^2/98$).
- [4] Ibid.: `einstein_direct/README.md` §§1–3; `einstein_direct/README_EN.md` §22.
- [5] Monographs No. 1–2 and No. 8–10 of the theorems12 cycle.
- [6] M. W. Choptuik. Phys. Rev. Lett. 70, 9 (1993) — source of the $\gamma_{\text{lit}} = 0.374$ comparison anchor.